

C-GEM: Plan for Data Management

Overview. This Data Management Plan has been prepared in accordance with guidelines from the NSF Division of Chemistry, “*Advice to PIs on Data Management Plans*,” and is consistent with the NSF Policy on Dissemination and Sharing of Research Results.

Data Sharing Across Teams. The C-GEM team will store, share, disseminate, and archive data following well-established, open, effective, secure, and timely processes. At the core of our Data Management strategy is GEM-NET, a set of web tools that support virtual teamwork *via* a Google Team Drive that provides unlimited storage for presentations, documents, and other data, including the underlying data storage for our custom data management apps. These custom apps include: **Strains**, which tracks the identity and transfer of strains and plasmids; **Protocols**, which documents our laboratory procedures; and **Molecules** (in progress), which houses information on the structures, synthesis, characterization, and team usage of C-GEM monomers. Future work will incorporate a “search by structure” capability to help users find monomers quickly even when the archive scales to thousands of entries. We will also develop a searchable archive of **GEM-NET Laboratory Notebooks** that document research and can be associated with specific records in our **Protocols**, **Strains**, and **Molecules** databases. All new team members will receive a GEM-NET orientation to help them become rapidly proficient in each of these tools.

1. C-GEM Research Products. **(A) Materials.** C-GEM scientists will produce new chemical compounds, new biological reagents, and new bacterial strains. **(B) Experimental Data.** Data will be collected from chemical synthesis procedures, biochemical, structural, and molecular biology experiments, and structural and functional characterization of C-GEM polymers. Such data include spectra, chromatograms, diffraction data, images, and results of properties analysis. Metadata will be generated describing the storage locations of chemical compounds, plasmids, and bacterial strains. We will also generate annotations of various sequence information (polymers, proteins, DNA). **(C) Software.** We will generate software including custom GEM-NET apps written in Python, algorithms for the Rosetta molecular modeling software suite, and simulation programs. **(D) Other Products.** We will generate experimental protocols and collect data on GEM-NET usage from component services. **(E) Education and Outreach Materials.** Our broader impacts and public engagement activities will generate educational materials (syllabi, lectures, demonstrations, hands-on activities) and online content (videos, blog posts, podcasts). **(F) Evaluation Data.** Our management and broader impact activities will generate assessments, narratives, and aggregated information about participants. Human subject protocols will be developed and submitted for IRB review/approval at each institution pending notice of award.

2. Data Format. Collected data will be formatted in several ways: (A) images as .tiff files; (B) experimental data as csv/Excel files or more specialized formats from the output of FACS, LC-MS, cryo-EM, etc.; and (C) laboratory protocols and notes, initially as pdfs or images scanned from lab notebooks, and then in a common GEM-NET Notebook format. Whenever possible, we will save data in machine-readable non-proprietary formats; in cases where that is not possible, we will provide images or extracted data to interested parties upon request.

3. Access to Data and Data Sharing Practices and Policies. **(A) Intellectual Property,** copyrights, and data will be administered in accordance with the policies of NSF and the participating institution(s) and codified in the C-GEM Collaboration Agreement. We expect all researchers to respect their responsibilities to the chemical and broader scientific communities and provide their results, data, and collections to other researchers upon request and in accordance with

all university guidelines. **(B) Materials** will be disseminated in accordance with participating institutional and NSF policies. Depending on the terms of those policies, materials may be transferred to others under Material Transfer Agreements (MTAs). **(C) Research Data** will be made available after the findings have been accepted for publication. Research products will be preserved for open access wherever possible. Structural data will be deposited in the appropriate database upon publication, following standard practice. **(D) Software.** Source code for custom GEM-NET apps will be distributed through GitHub. Rosetta algorithms will be made available on the Rosetta Commons. The SCA4RNA algorithm will be made available on the C-GEM website. **(E) Publications.** C-GEM investigators will promptly prepare and submit manuscripts resulting from NSF support, assign authorship to accurately reflect contributions, and acknowledge funding support in accordance with agency policies. Our **Code of Conduct** encourages submission of pre-prints prior to publication. **(F) EteRNA.** Citizen scientists who engage in C-GEM×EteRNA design challenges will access project data in a convenient, searchable format through the EteRNA lab visualizer after registration (free of charge) with the EteRNA project (<http://eternagame.org>). Under the EteRNA User Agreement, players relinquish any IP claims and consent to sharing their designs with collaborators. The EteRNA project (and individual players, when appropriate) will receive authorship and/or acknowledgement on C-GEM publications following established norms.

4. Policies for Re-use, Re-distribution, and Production of Derivatives. The Director has overall responsibility for both data management and dissemination. C-GEM Investigators will share with other researchers, at no more than incremental cost and within reasonable time, the primary data, samples, physical collections, software, curriculum materials, and other supporting materials created or gathered in the course of NSF-funded work. Each collaborator will sign a Collaboration Agreement pledging to abide by the terms of this Data Management Plan. **(A) Sensitive data** will be made available to the Director, Center staff, and select members of the External Advisory Board with relevant expertise. Privacy, confidentiality, and security of personal data collected for evaluation will be ensured *via* the following standard practices: collecting data anonymously (through Google Forms); refraining from collection of demographic data that could render individuals identifiable; and storing sensitive data on a separate password-protected GEM-NET Team Drive with limited access. **(B) Materials for general access** will be shared on the C-GEM website (educational materials, blog posts, resources) under a CC BY-NC-SA 3.0 license that allows users to share and adapt material for noncommercial use, requires attribution, and requires any new material created to be distributed under the same terms. Podcasts are disseminated through Yale University on Apple Podcasts and SoundCloud (CC BY-NC-SA 3.0 license).

5. Archival Storage Standards. Preliminary data will be stored on office computers and local data systems accessible from workstations associated with each C-GEM Investigator. Data in non-electronic form will be converted to digital form for preservation. Final responsibility for data archiving and preservation lies with each Investigator. At the conclusion of the grant, data can be maintained indefinitely on local backups and also, for as long as C-GEM exists as an educational entity, on the Team Drive. **(A) Timing.** Data will be archived within 1 month of validation or upon publication. **(B) Method of Archiving.** Data will be uploaded to a “Data” Team Drive, the contents of which will be backed up weekly to a local server at UC-Berkeley. **(C) Preservation of Access.** Access to Team Drive and GEM-NET apps is provisioned to individual google accounts. We will use Google Groups for access control and will create an “alumni” group for former C-GEM members with read-only permissions. If necessary, we can create GEM-NET.net Google accounts for departing members.